CAUSA

“Answers you can check”

Track 02: The Agent That Can Explain Why

Team MARS · GenAI Works Open Agent Hackathon 2026

The problem: “why” has no audit trail

Questions analysts get

“Why did Target’s revenue growth lag Walmart’s from FY2022 to FY2024?”
“Is Ford’s slowdown company-specific or sector-wide?”

Why it is hard

Evidence is fragmented: SEC filings, industry classifications and macro statistics live in different places, and are stitched together by hand.

What they end up with

A slide with no audit trail. Nobody can check which numbers came from where.

Target users

Equity, credit and FP&A analysts who must explain movements across filings, peers and macro data.

The solution: an analyst agent that shows its work

1

Ground

Resolve companies to real CIK, ticker and SIC. Unknown names are refused, not invented.

2

Hypothesise

Propose competing causes: company-specific, sector-wide or macro.

3

Test

Read-only SQL through Zetaris across EDGAR, SIC and World Bank data.

4

Critique

A critic pass checks every number in every claim against the rows it cites.

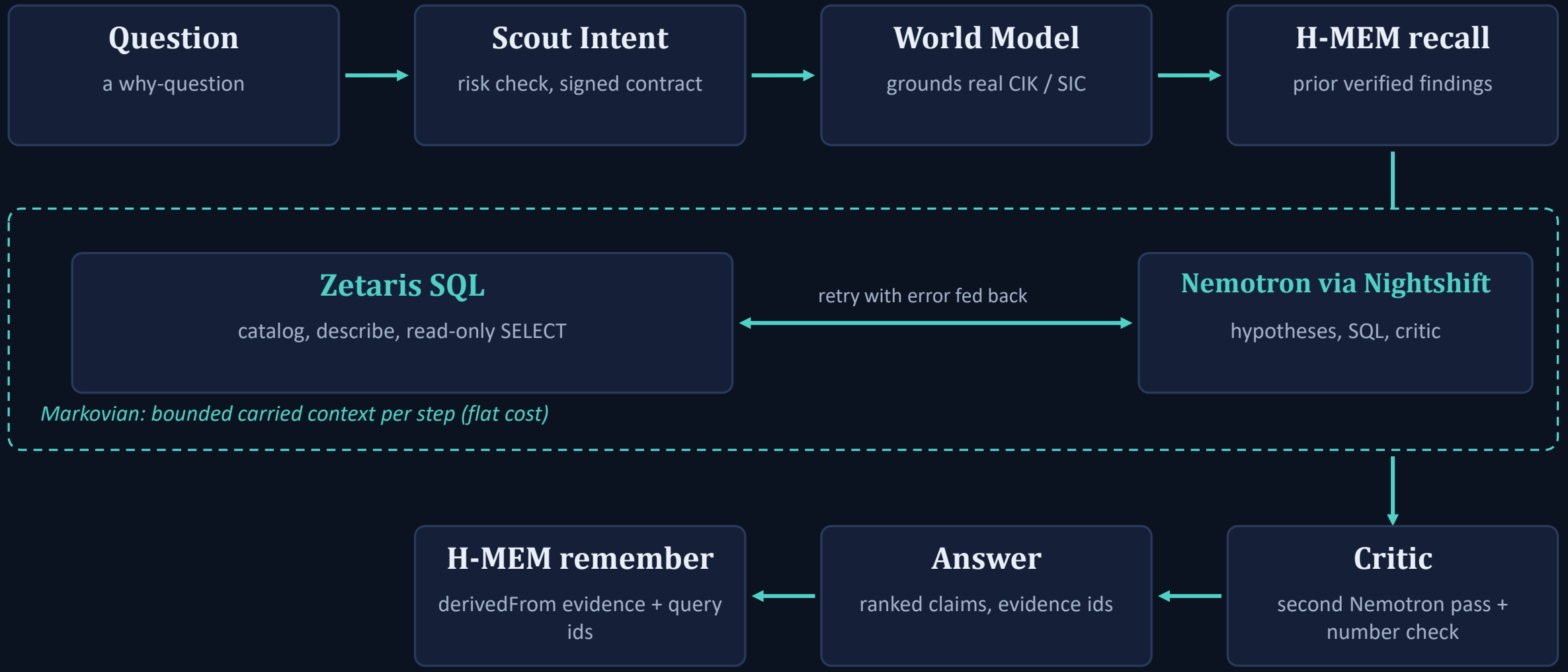
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Answer

Ranked causes with confidence, each linked to the exact query, rows and source.

Verified findings are remembered with provenance, so the next related question starts warm and can be audited.

Agent workflow and architecture



Streams live to the web UI over Server-Sent Events

Technical stack

Data

Zetaris Lightning SQL over REST, federated sources + Unified Semantic Layer

Power

Meterless Nightshift, OpenAI-compatible runtime (primary AI provider)

Model

NVIDIA Nemotron models, served through Nightshift

Memory

Meterless H-MEM

Grounding

Meterless World Model

Context

Meterless Markovian

Intent and safety

Meterless Scout Intent

UI and API

Vite + React + TypeScript; Node 22 + Hono, SSE

Built with Claude Code. Cursor was not used.

How Zetaris and Meterless were used

Technology	What it does in Causa
Zetaris	Every query: Lightning SQL over REST across EDGAR XBRL, SIC hierarchy, World Bank, plus a Unified Semantic Layer
Meterless Nightshift	Primary AI provider; the model that handled each call is recorded and shown
Meterless H-MEM	Remembers verified findings with provenance; recall, audit trail, feedback
Meterless World Model	Grounds companies to real CIK / ticker / SIC; refuses unknown names
Meterless Markovian	Keeps each reasoning step’s carried context bounded (flat cost per step)
Meterless Scout Intent	Intent, risk check, HMAC-signed contract with enforced tools and budgets
NVIDIA Nemotron	Planner, SQL investigator and critic, served through Nightshift
Cursor	Not used (built with Claude Code)

What was unique: a verification chain

1 Scout Intent

decides what the agent may do

2 World Model

decides what exists

3 Zetaris

the only source of facts

4 Nemotron

reasons over the evidence

5 Critic

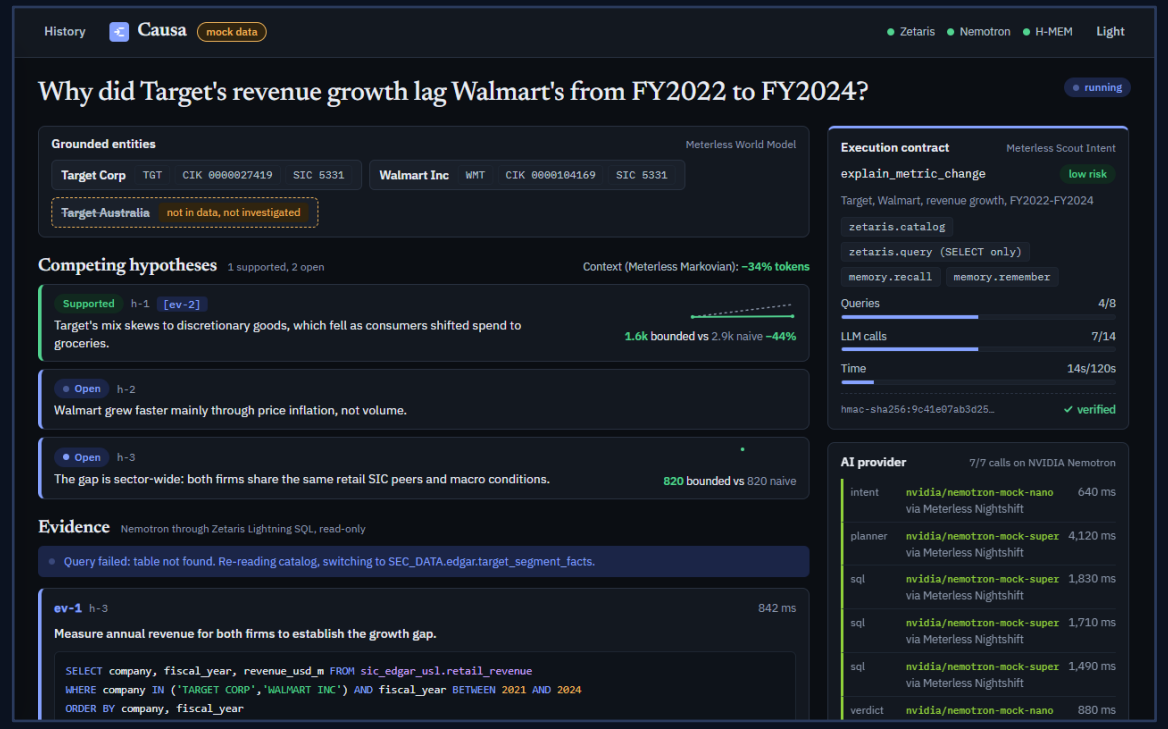
refuses any number not in the evidence rows

6 H-MEM

remembers only what survived, with provenance

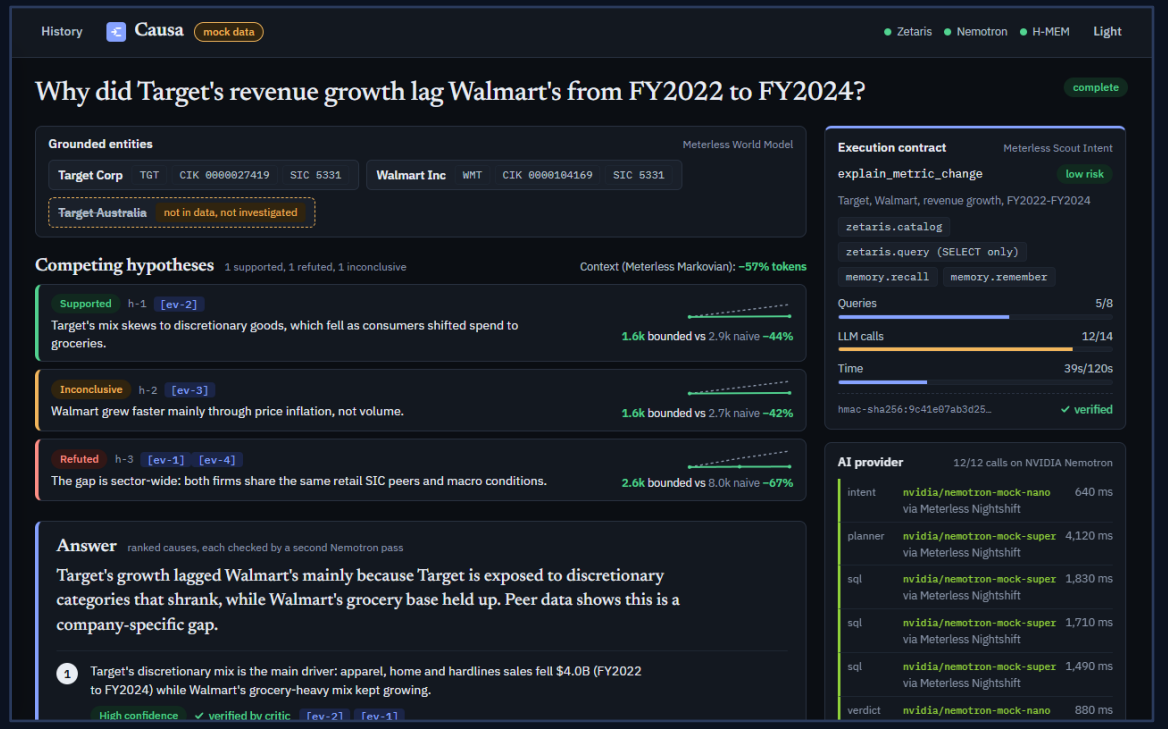
Each step constrains the next; nothing reaches memory unchecked.

Product visuals: the investigation



Contract, entities, hypothesis board

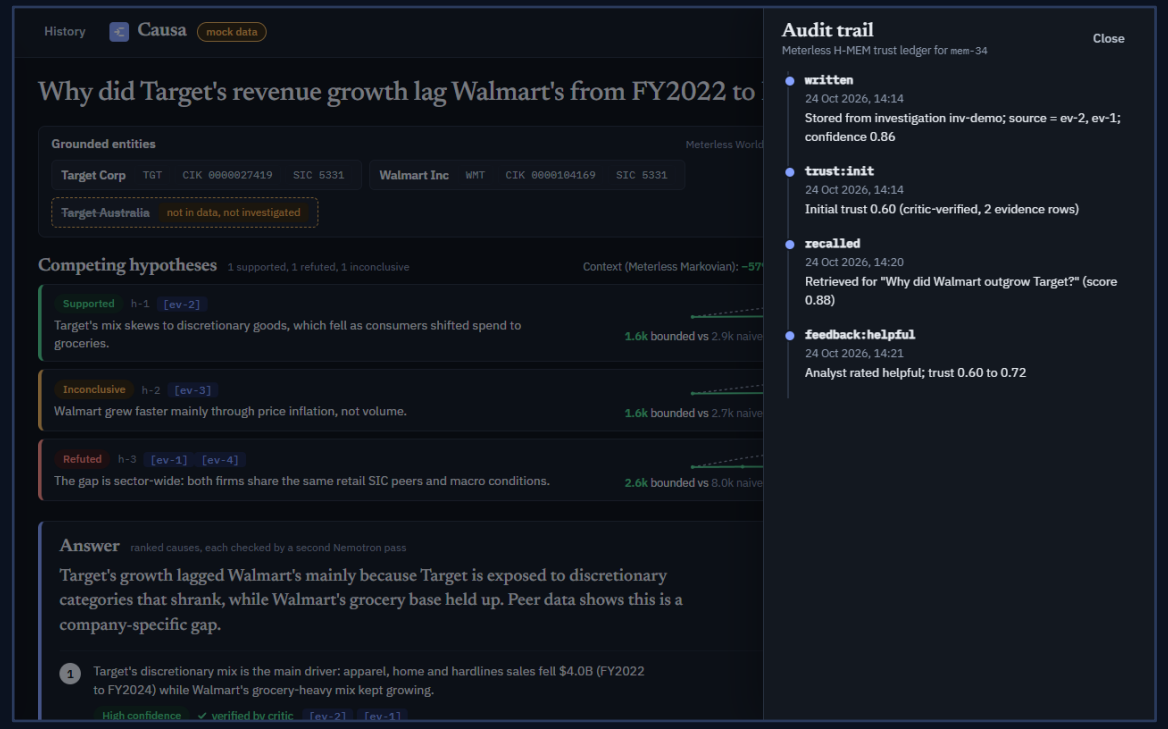
UI preview



Ranked answer, checked by a critic

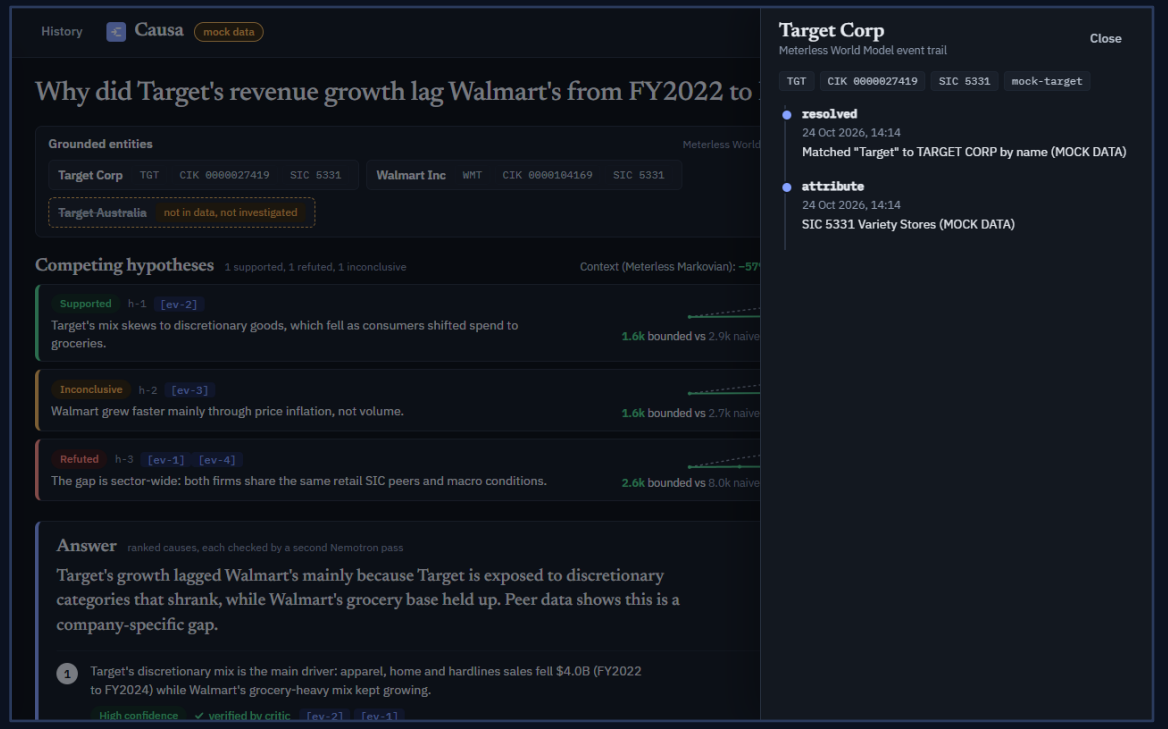
UI preview

Product visuals: audit and entity trail



H-MEM audit trail

UI preview



World Model entity trail

UI preview

Future enhancements

More sources

FUTURE

Earnings-call transcripts and internal ERP, via Zetaris connectors

Overnight investigations

FUTURE

Scheduled Nightshift runs that pre-answer recurring questions

Analyst feedback loop

FUTURE

Analyst feedback written back into H-MEM

Team-shared memory

FUTURE

Verified findings shared across an analyst team

Evaluation set

FUTURE

Historical “why” questions with known answers to measure the agent

Causa

“Answers you can check”

https://github.com/ShayanMuhammad-CS/causa

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